

CS 486/686

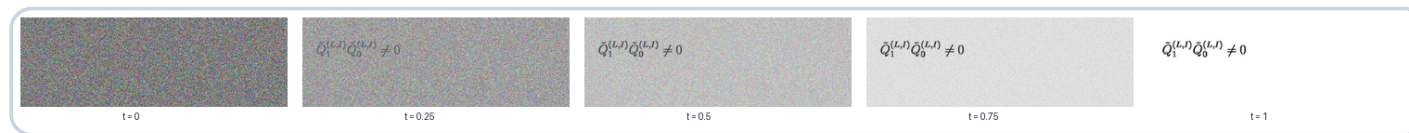
Dissecting Diffusion and World Models

Yuntian Deng

Lecture 24

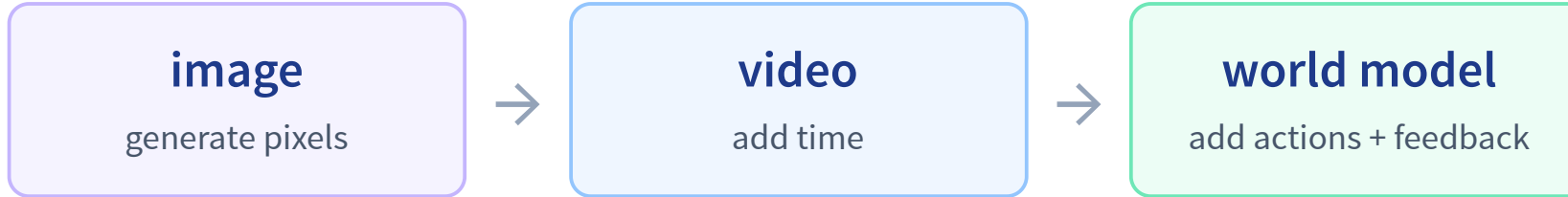
From noise to video to an interactive world

From noise to an image



[Deng, Kojima & Rush, “Markup-to-Image Diffusion Models with Scheduled Sampling,” ICLR 2023.](#)

Image → video → interactive computer



PART I

Image generation

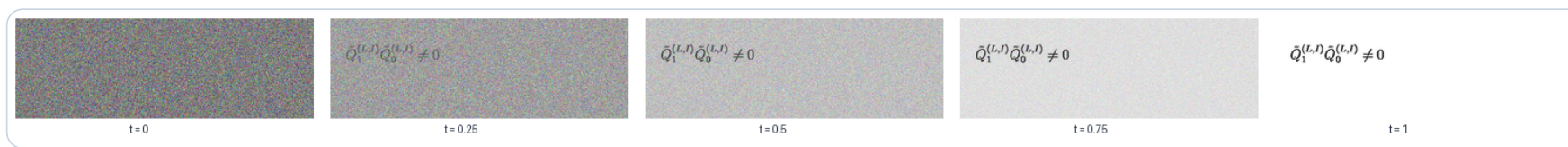
Learn a path from noise to data.

Start with noise and an image



Generation learns how to travel from the left endpoint to the right.

Connect them with a path INTERACTIVE



$$z_t = (1 - t)z_{\text{noise}} + tz_{\text{data}}$$

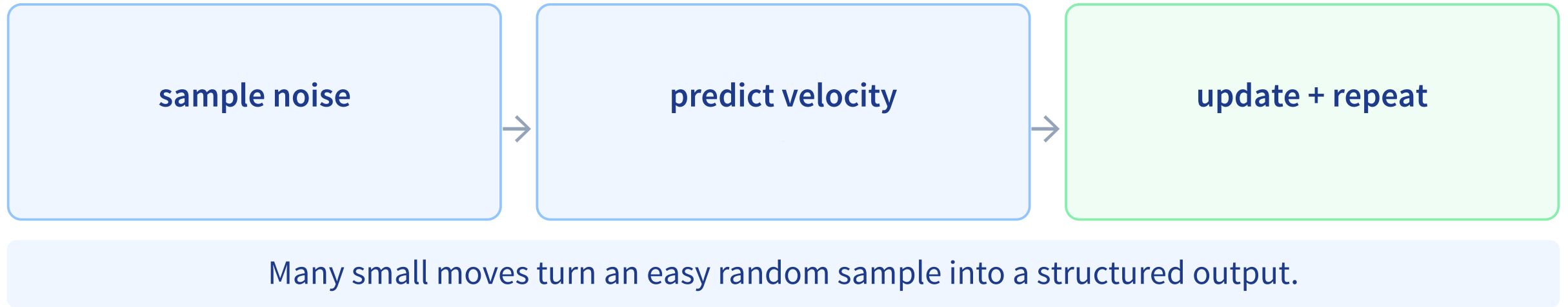
Velocity is the slope of the path

$$z_t = (1 - t)z_{\text{noise}} + tz_{\text{data}}$$

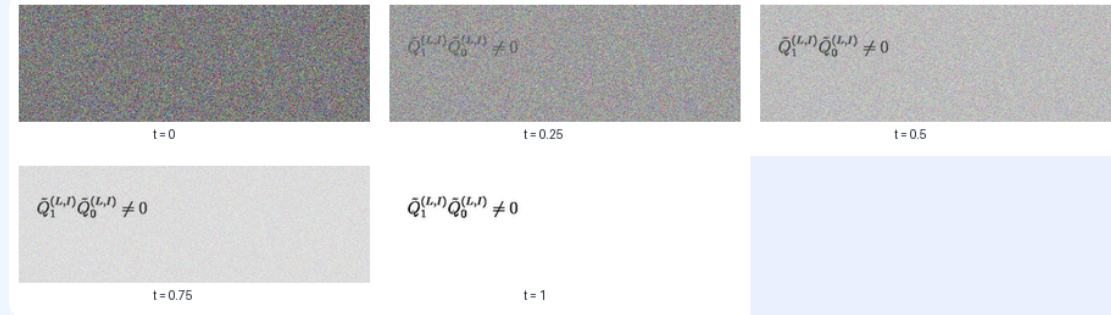
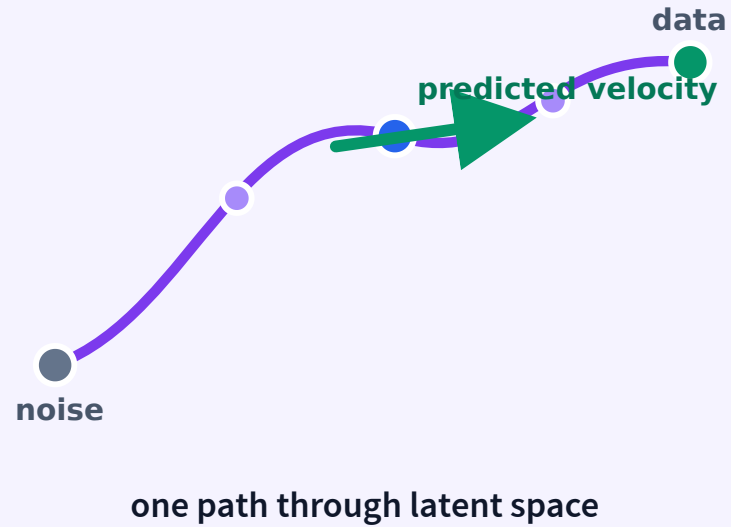
↓ differentiate with respect to t

$$v^{\star} = \frac{dz_t}{dt} = z_{\text{data}} - z_{\text{noise}}$$

Generation follows those arrows repeatedly



Each velocity prediction advances the sample



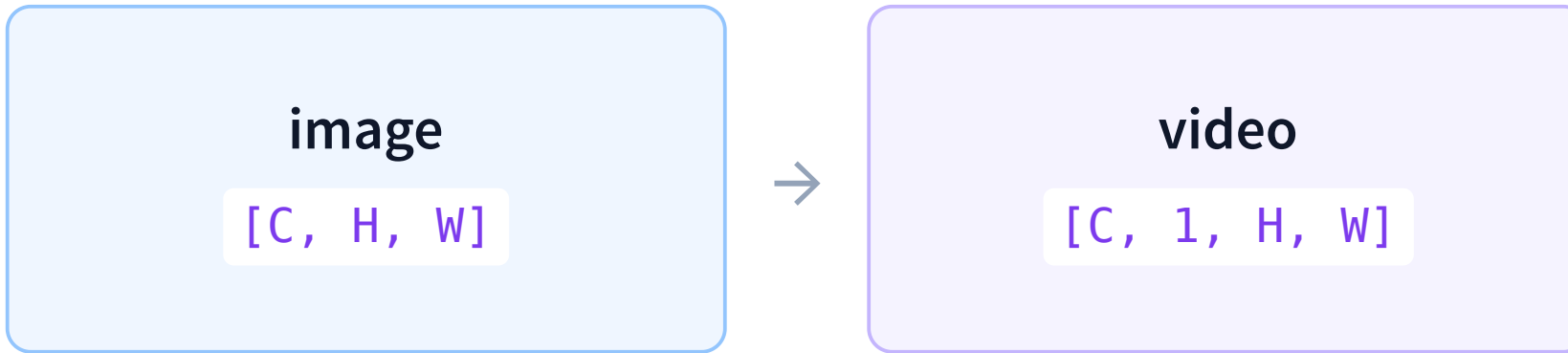
the corresponding output becomes cleaner

PART II

Video generation

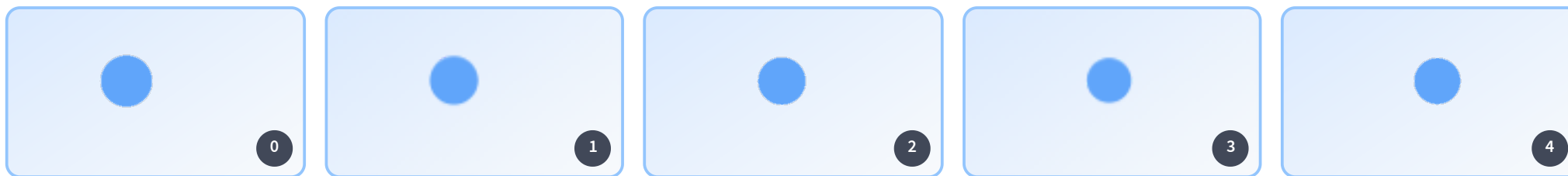
Generate many consistent frames—not just one image.

An image is a one-frame video



Add one axis: time.

Video adds time



same object

smooth motion

consistent scene

Five seconds is about 80 frames

$$5 \text{ seconds} \times 16 \text{ frames/second} \approx 80 \text{ frames}$$

Five seconds already means 96 million values

$$80 \times 480 \times 832 \times 3 = 95,846,400$$

$$\boxed{80 \text{ frames}} \times \boxed{480 \times 832 \text{ pixels}} \times \boxed{\text{RGB}}$$

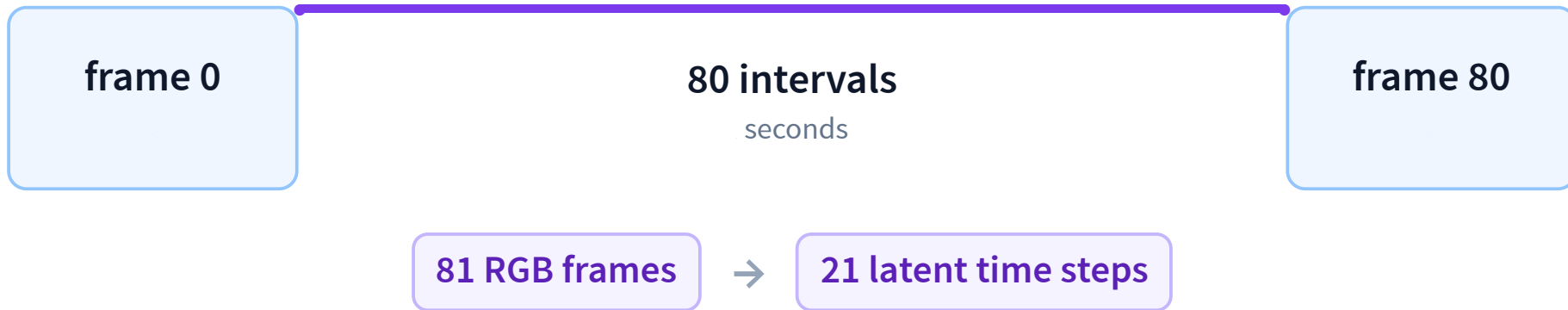
Compress first; run the expensive transformer in a smaller space.

CASE STUDY

Wan2.1-T2V-1.3B

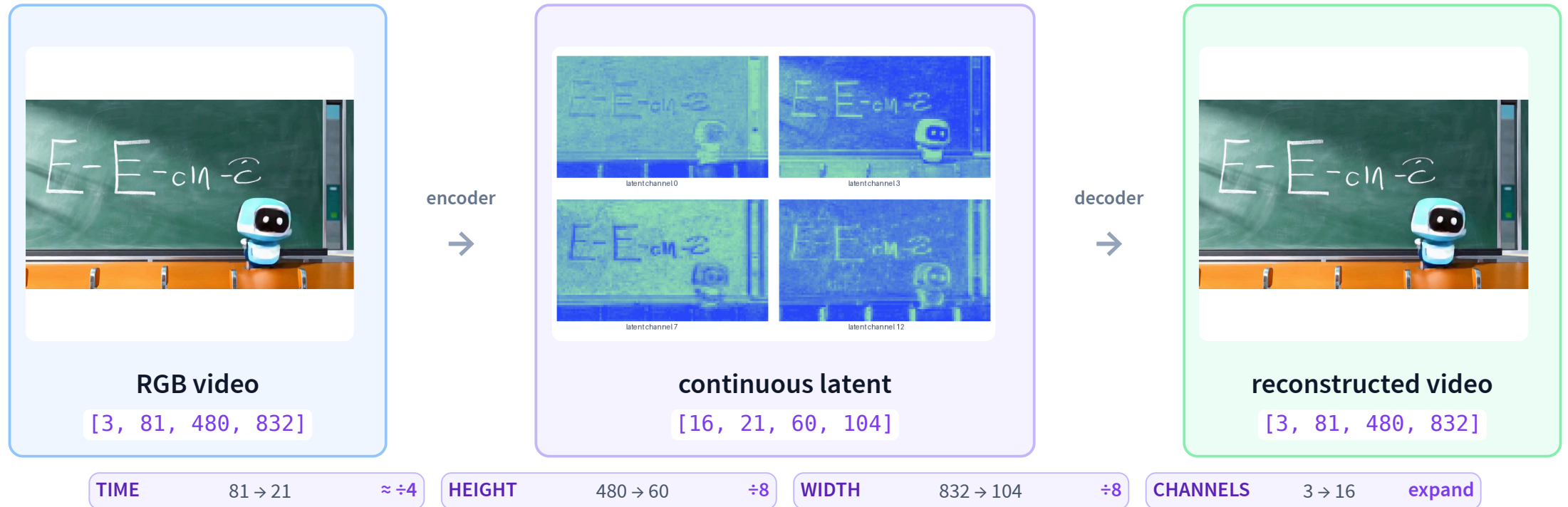
An open text-to-video model built from
a video VAE, a DiT, and a text encoder.

Wan's default clip has 81 frames



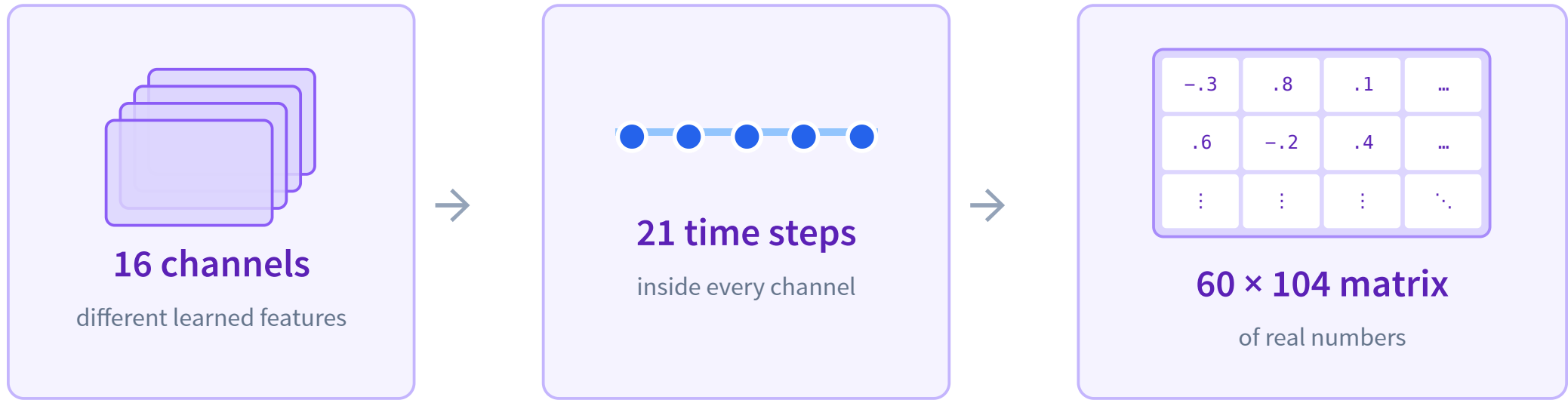
Other supported lengths follow $4k + 1$, such as 33, 49, or 81 frames.

The Wan-VAE compresses the video



Team Wan et al., [“Wan: Open and Advanced Large-Scale Video Generative Models,” 2025; example provenance.](#)

A latent video is a nested array



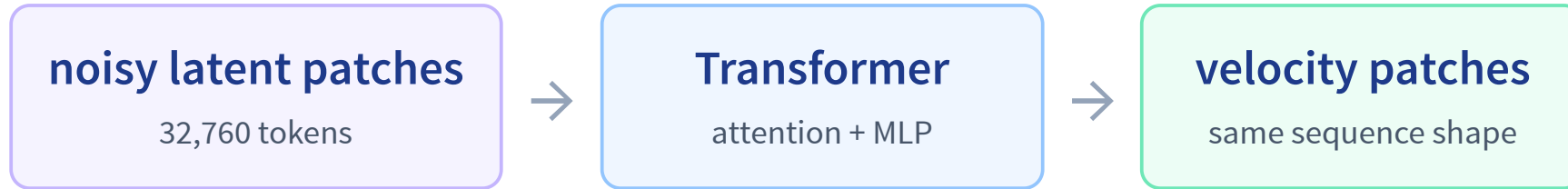
$$[16, 21, 60, 104] = z[\text{channel}, \text{time}, \text{row}, \text{column}]$$

Patchify the latent—just like an image encoder



One token begins as $16 \text{ channels} \times 1 \text{ time step} \times 2 \times 2 \text{ spatial cells} = 64 \text{ values}$.

A DiT is a transformer for noisy latent patches



Peebles & Xie, [“Scalable Diffusion Models with Transformers,” ICCV 2023.](#)

Wan2.1-T2V-1.3B at a glance

Wan2.1

T2V-1.3B

Wan-AI/Wan2.1-T2V-1.3B

1.3B

DiT parameters

30

DiT blocks

1536

hidden width

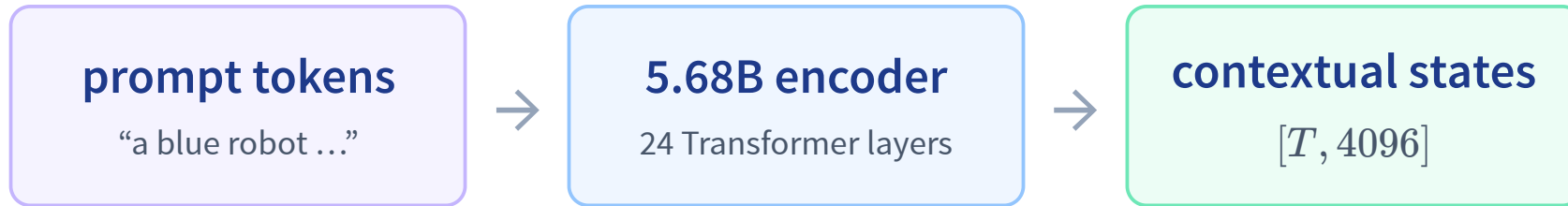
480p

target resolution

Team Wan et al., [“Wan: Open and Advanced Large-Scale Video Generative Models,” 2025](#); official checkpoint configuration.

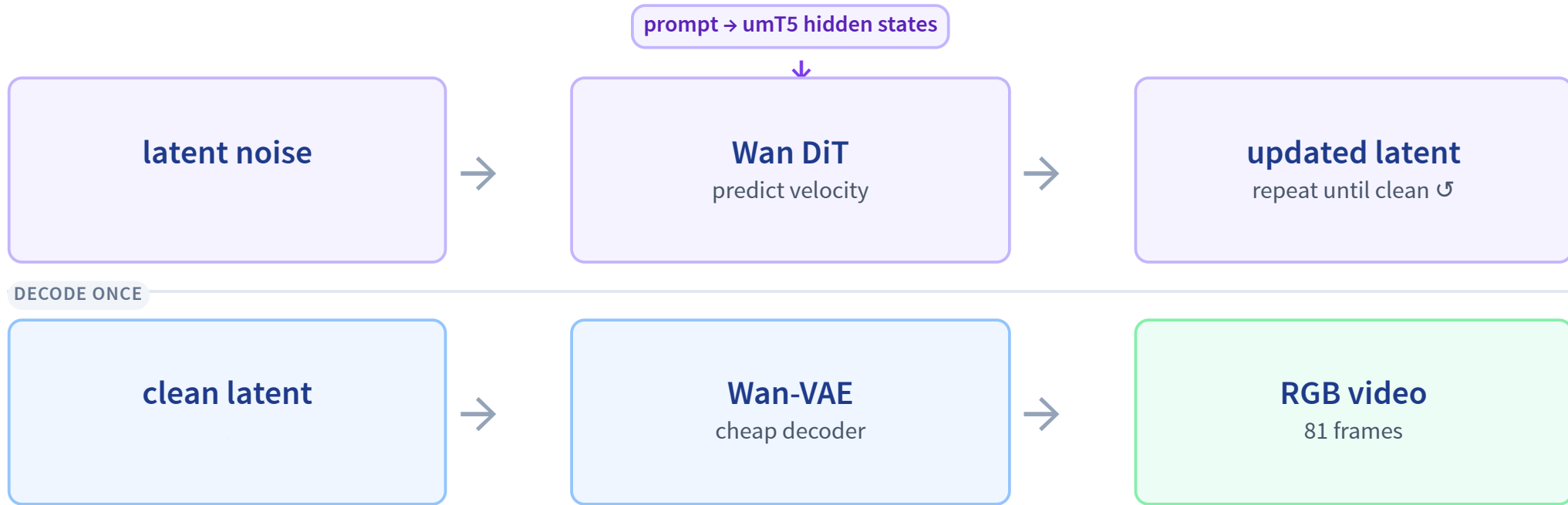
Wan uses only the umT5-XXL encoder

umT5-XXL is a ~13B encoder-decoder Transformer; Wan keeps the 5.68B encoder and discards the decoder.

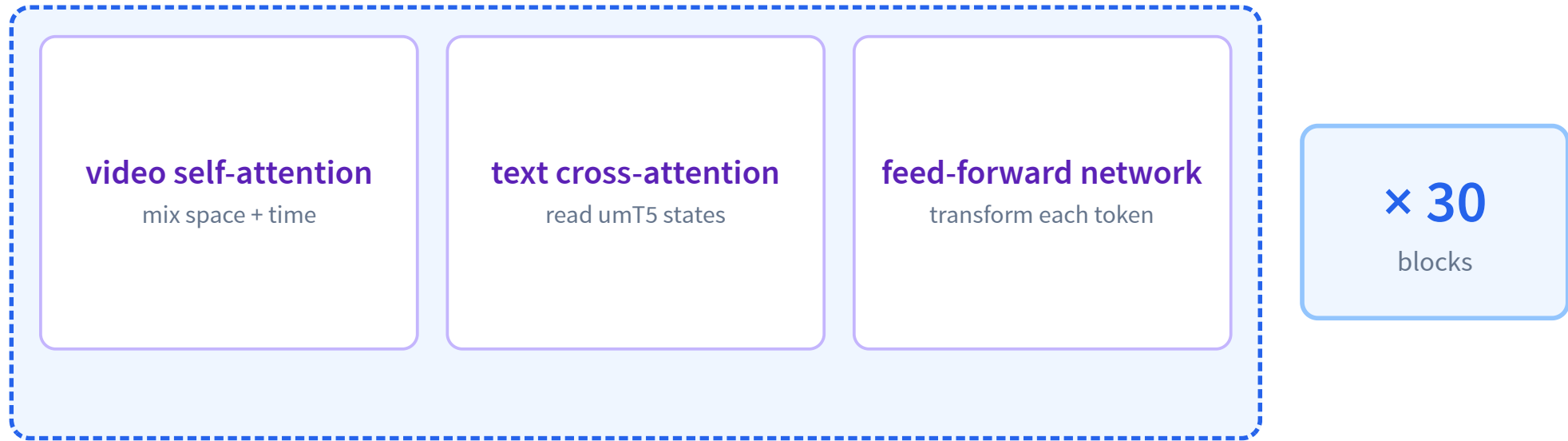


Chung et al., [“UniMax: Fairer and More Effective Language Sampling for Large-Scale Multilingual Pretraining,” 2023.](#)

Prompt in; video out

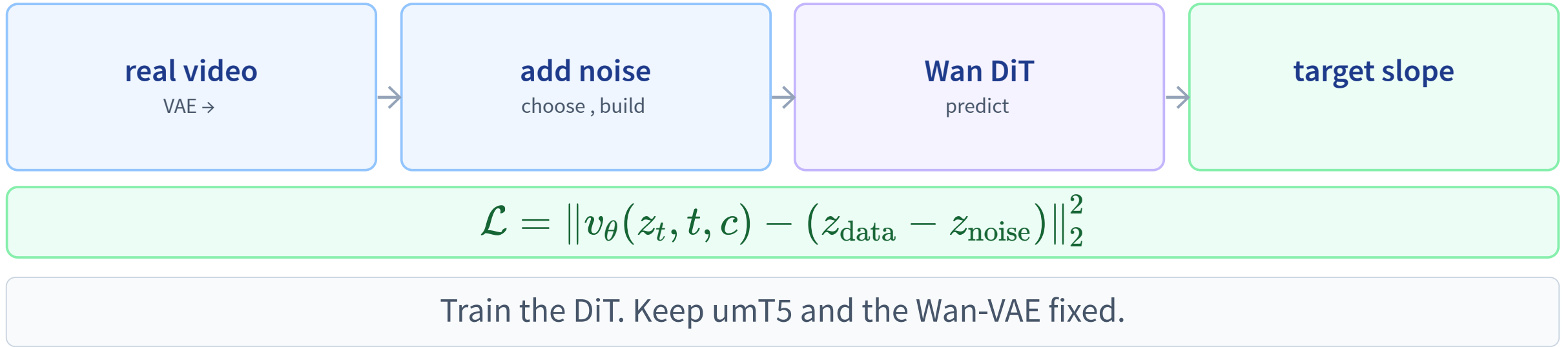


A Wan block mixes video, text, and time



The timestep modulates each block, telling it how noisy the current latent is.

Train the DiT to predict velocity

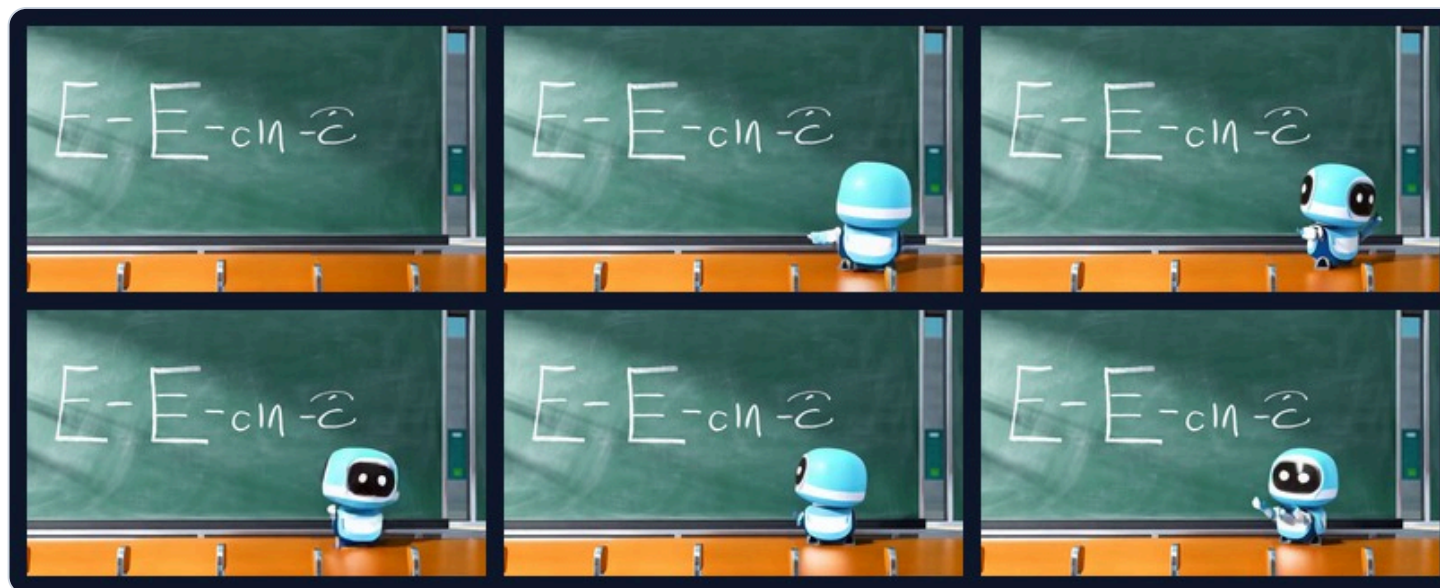


The generated result

RECORDED MODEL OUTPUT

PROMPT

A small blue robot writes $E = mc^2$ on a blackboard in a sunlit university classroom, then turns toward the camera and waves, fixed camera, coherent motion, clean cinematic lighting.



Generated with [Wan2.1-T2V-1.3B](#); [reproducibility record](#).

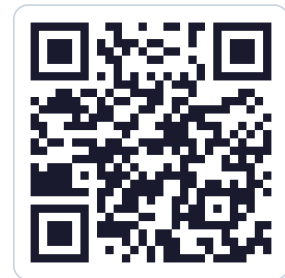
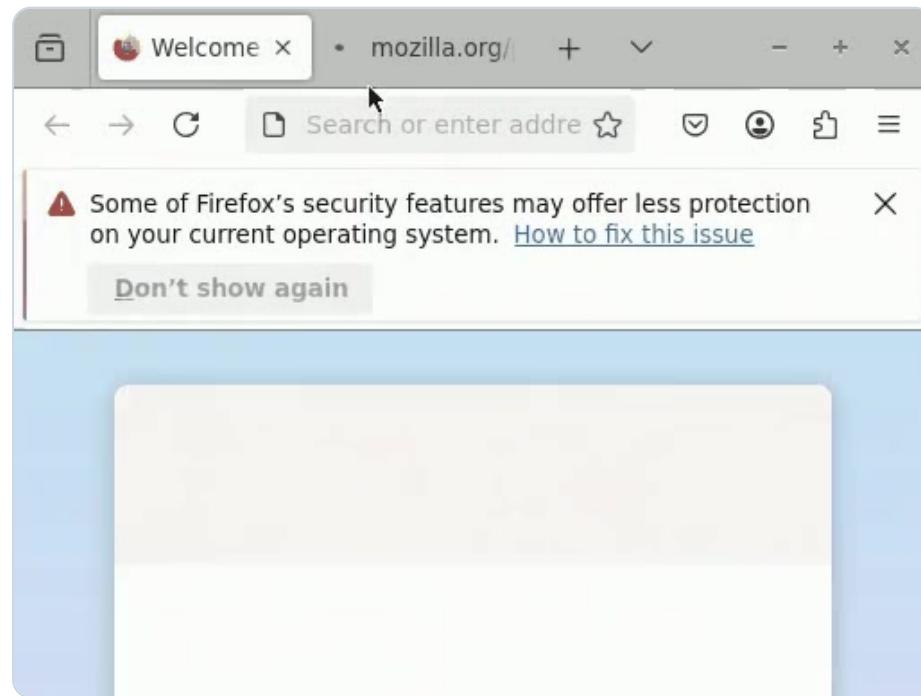
PART III

From video generation to world modeling

A fixed video cannot react. A world model observes actions and generates what happens next.

What if the whole computer were a neural network?

Mouse and keyboard go in. The next screen is generated.



neural-os.com

Rivard et al., [“NeuralOS: Towards Simulating Operating Systems via Neural Generative Models,” ICLR 2026.](#)

A video generation model is not a world model

video generator

prompt once

fixed clip

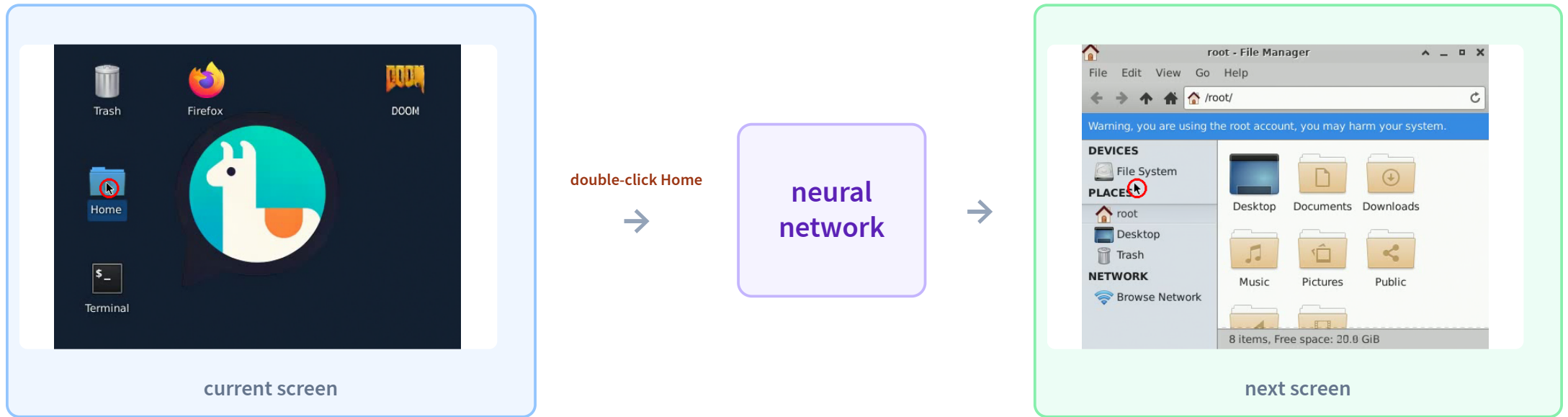
world model

observe each action

predict → feed back ↺

A generated video is static. You cannot interact with it like a video game.

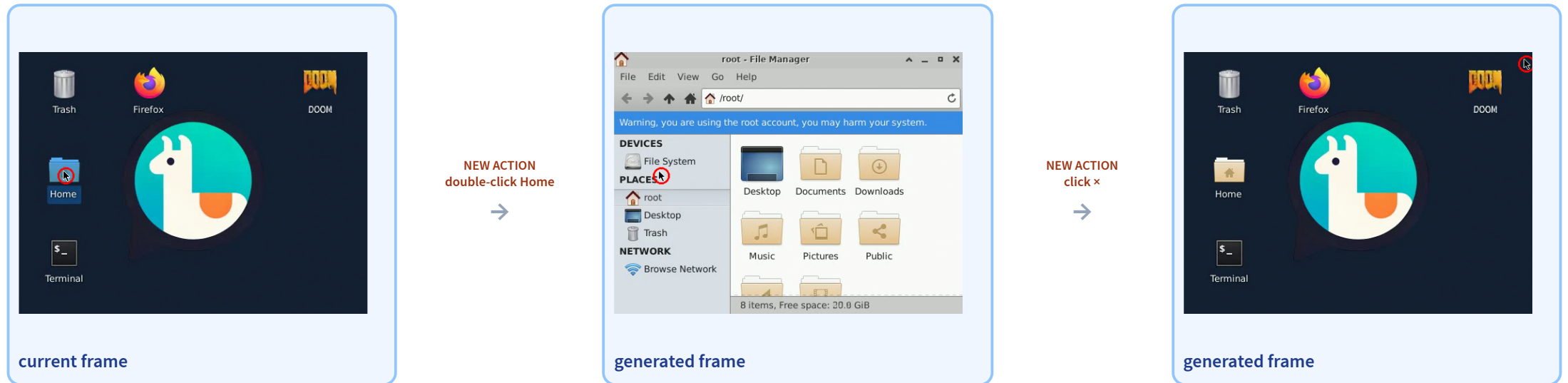
NeuralOS predicts the next screen



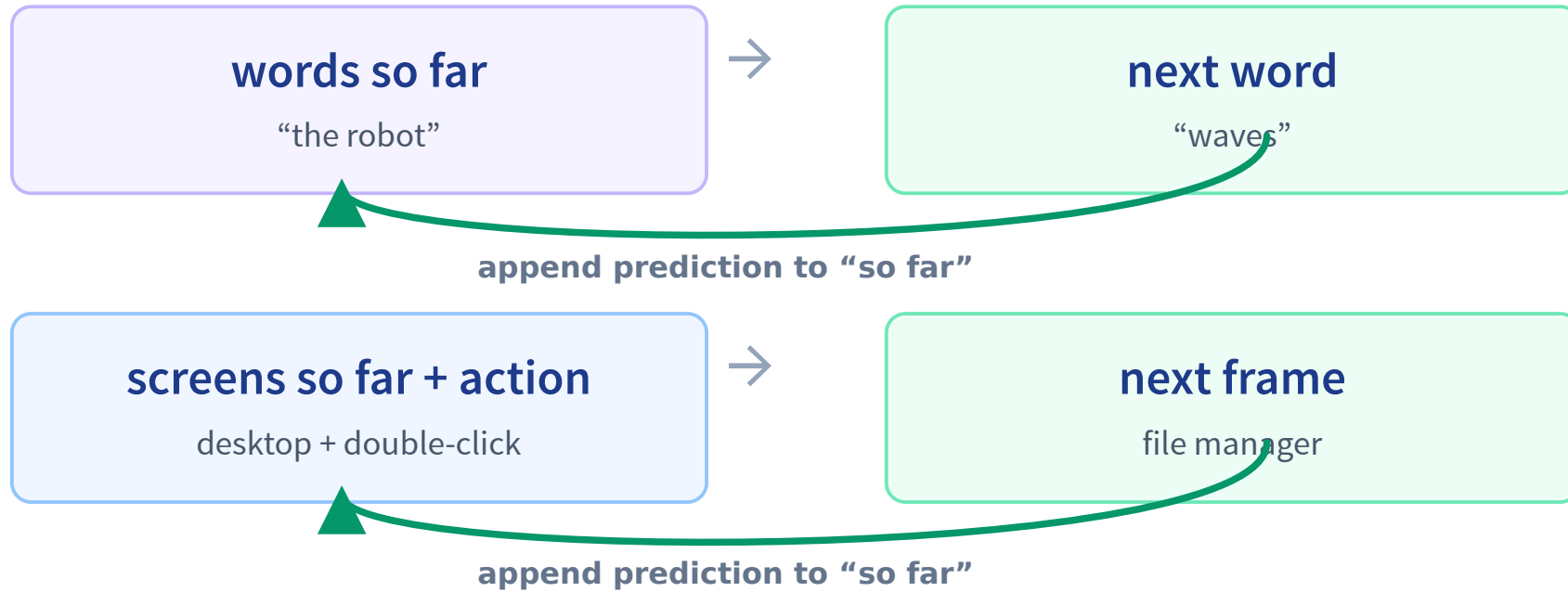
(screen history, action) → next screen

A new action generates a new frame INTERACTIVE

Click an action. The model generates the next screen—there is no autoplay.

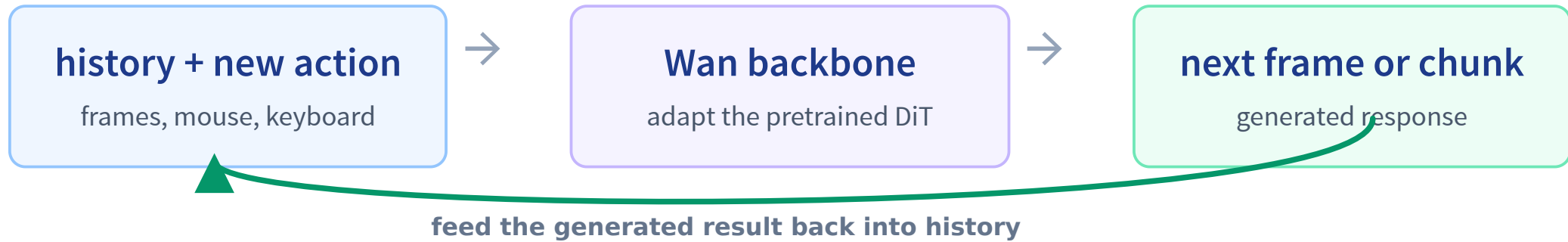


Next word prediction → next frame prediction



Wan can be adapted to predict what happens next

Why reuse Wan? Video pretraining has already learned useful priors about appearance, motion, and dynamics.



NeuralOS uses a different backbone. The shared formulation is repeated next-frame prediction.

Zhao et al., [“minWM,” 2026](#); He et al., [“Matrix-Game 2.0,” 2025](#).

Before you go: deadlines and office hour

OFFICE HOUR TODAY

4–5 PM

MC 4021 · this classroom

ASSIGNMENT 3

Tue Aug 4 · 11:59 PM

Waterloo time

CS 686 FINAL PROJECT

Wed Aug 5 · 11:59 PM

Report + code submission

Final exam—and good luck

SATURDAY AUG 8

7:30–10:00 PM

PAC 5

BRING

Non-programmable calculator

One double-sided letter-size notes sheet

This is our last class. Good luck!